

Chapter 5: Towards Green Energy

Page 47 – Easy Exam-Oriented Notes

What is Energy?

- **Energy** is the **capacity to do work**.
 - Without energy, no work can be done.
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Types of Energy (Based on Source)

1. **Renewable Energy** – Can be reused again and again.
 - Examples: Solar energy, Wind energy, Water (Hydel) energy
 2. **Non-renewable Energy** – Limited and get exhausted.
 - Examples: Coal, Petroleum, Natural gas
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Forms of Energy

Energy exists in different forms:

- **Mechanical Energy** – Energy of motion or position
 - **Heat Energy** – Energy due to temperature
 - **Light Energy** – Energy from sun, bulb, etc.
 - **Sound Energy** – Energy due to vibrations
 - **Electrical Energy** – Energy due to flow of electric current
 - **Chemical Energy** – Energy stored in food, fuel, batteries
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Use of Energy in Day-to-Day Life

Daily Work	Form of Energy Used
Walking, running	Mechanical energy
Cooking food	Heat energy
Watching TV	Electrical + Light energy

Daily Work	Form of Energy Used
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Listening to music	Sound energy
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Studying with lamp	Electrical + Light energy
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Use of Electrical Energy in Daily Life

Electrical energy is used for:

- Lighting (bulb, tube light)
 - Running fans, TV, refrigerator
 - Charging mobile phones
 - Operating machines and computers
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How is Electrical Energy Produced?

- Electrical energy is produced by **electric generators**.
 - It is based on the principle of **Electromagnetic Induction**.
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Principle of Electromagnetic Induction

- Discovered by **Michael Faraday**.
 - **When the magnetic field around a conductor changes, electric current is produced in it.**
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Ways to Change Magnetic Field

1. Rotating magnet near a stationary conductor
2. Moving conductor in a stationary magnetic field

 In both cases, **potential difference** is created and electricity is generated.

Electric Generator

- A machine that converts **mechanical energy into electrical energy**.
 - Works on the principle of electromagnetic induction.
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Importance of Energy in Modern Life

- Energy is a **basic need** like food, clothing and shelter.
 - We need energy for industries, transport, communication and homes.
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Conversion of Energy

- Energy can be converted from one form to another.
 - Example:
 - Mechanical → Electrical (Generator)
 - Electrical → Light (Bulb)
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Keywords for Exam

- Energy
 - Renewable energy
 - Non-renewable energy
 - Electromagnetic induction
 - Electric generator
 - Potential difference
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5.1 Electromagnetic Induction (Recap)

- When a **magnetic field around a conductor changes**, an **induced emf (electric current)** is produced.
- This phenomenon is called **Electromagnetic Induction**.
- It is the basic principle of **electric generator**.

Important parts involved:

- Magnet / Electromagnet
 - Coil (conductor)
 - Rotational motion
 - Voltmeter (to show induced emf)
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5.2 Steam Turbine

- A **turbine** is a machine used to rotate the magnet of a generator.
- It has **blades** fixed on a wheel.
- When **high velocity steam or fluid** strikes the blades, the turbine rotates.
- Rotation occurs due to **kinetic energy** of steam or fluid.

Main Parts of Steam Turbine:

- Nozzle – directs steam
 - Blades – rotate due to steam
 - Wheel and Shaft – transfer motion to generator
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5.3 Electric Generator (Schematic Idea)

- Generator converts **mechanical energy into electrical energy**.
- It consists of:
 - Rotating magnet / electromagnet
 - Coil
 - Shaft

 Rotation of magnet causes change in magnetic field → electricity is generated.

5.4 General Flow Chart of Electrical Energy Generation

Energy Source → Turbine → Generator → Electrical Energy

- Energy source is required to rotate the turbine.
 - Turbine rotates the generator.
 - Generator produces electrical energy.
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Types of Power Generating Stations

- Depending on the **energy source used to rotate turbine**, power stations are of different types:
 - Thermal power station
 - Hydel power station
 - Nuclear power station
 - Wind power station
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Thermal Energy Based Power Station

- Uses **coal** as fuel.
 - Coal is burned to produce **thermal energy**.
 - Water is heated in a **boiler** to produce **high temperature and high pressure steam**.
 - Steam rotates the turbine.
 - Turbine rotates the generator.
 - Electrical energy is produced.
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Flow Chart: Thermal Power Station

Coal → Boiler → Steam → Steam Turbine → Generator → Electrical Energy

- After rotation, steam is cooled and converted back into water.
 - Water is **re-circulated** to the boiler.
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Important Exam Points

- Turbine converts **kinetic energy of steam** into mechanical energy.
 - Generator converts **mechanical energy into electrical energy**.
 - Steam power station is an example of **non-renewable energy source**.
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Keywords for Exam

- Electromagnetic induction
- Turbine
- Steam turbine

- Generator
 - Thermal power station
 - Boiler
 - Kinetic energy
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Towers Seen in a Thermal Power Station

If you observe a thermal power plant, you will see **two types of towers**:

1. **Stack (Chimney)** – Releases combustion gases into the atmosphere.
 2. **Cooling Tower** – Releases heat from hot water in the form of vapour and hot air.
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Arrangement in Thermal Power Station

- **Boiler** – Burns coal and converts water into steam.
 - **Turbine** – Rotated by high pressure steam.
 - **Generator** – Produces electrical energy.
 - **Condenser** – Converts steam back into water.
 - **Cooling Tower** – Cools the water used in condenser.
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Working of Thermal Power Plant

- Coal is burned in the boiler.
 - Combustion gases go out through **high stack**.
 - Steam at high temperature and pressure rotates the turbine.
 - After rotation, steam loses heat and pressure.
 - Steam is cooled in the **condenser** and converted into water.
 - Cooling water absorbs heat and releases it through **cooling tower**.
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Why It Is Called Thermal Power Plant?

- Because **thermal energy (heat)** is used to generate electrical energy.
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Energy Transformation in Thermal Power Plant

Chemical energy in coal → Thermal energy → Kinetic energy of steam → Kinetic energy of turbine → Electrical energy

Let's Think – Important Answers

1. Why energy in coal is called chemical energy?

- Because energy is stored in coal due to chemical bonds.

2. Why steam is used to rotate turbine?

- Steam has **high pressure and high kinetic energy** which can rotate turbines efficiently.
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Problems of Thermal Power Plants

1. **Air pollution** due to gases like CO₂, SO₂, NO₂.
 2. **Soot particles** cause respiratory diseases.
 3. **Coal is limited** (non-renewable resource).
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Nuclear Energy Based Power Plant

Basic Principle

- Uses energy released by **nuclear fission** of Uranium or Plutonium.
 - Steam turbine is used similar to thermal power plant.
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Working of Nuclear Power Plant

- Nuclear fission releases heat energy.
- Heat converts water into steam.
- Steam rotates turbine.
- Turbine rotates generator.
- Electrical energy is produced.

Nuclear Fission (Easy Explanation)

- Neutron hits **Uranium-235**.
- It becomes **Uranium-236 (unstable)**.
- It splits into **Barium + Krypton + 3 neutrons**.
- About **200 MeV energy** is released.
- Released neutrons cause further fission (chain reaction).

Energy Transformation in Nuclear Power Plant

Nuclear energy → Thermal energy → Kinetic energy of steam → Kinetic energy of turbine → Electrical energy

Nuclear Fission & Nuclear Power Plant

Nuclear Fission (Chain Reaction)

- When **Uranium-235** absorbs a neutron, it splits into **Barium + Krypton**.
- **3 neutrons + huge energy (≈200 MeV)** are released.
- These neutrons cause further fission → **chain reaction**.
- In nuclear power plants, this chain reaction is **controlled**.

Advantages of Nuclear Power Plant

- Does not use fossil fuels like coal.
- No air pollution during electricity generation.
- Produces large amount of energy from small fuel.

Problems of Nuclear Power Plant

1. **Radioactive nuclear waste** – disposal is difficult.
 2. **Nuclear accidents** can be very dangerous.
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Comparison: Thermal vs Nuclear Power Plant

Similarities:

- Both use **steam turbine** and **generator**.
- Both convert thermal energy into electrical energy.

Differences:

- Thermal: Uses coal | Nuclear: Uses uranium/plutonium
 - Thermal: Air pollution | Nuclear: Radiation risk
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Power Plant Based on Natural Gas

Working

- Natural gas burns in **combustion chamber**.
- Hot gases at high pressure rotate **gas turbine**.
- Turbine rotates generator → electricity produced.

Energy Transformation

Chemical energy (natural gas) → Kinetic energy of gas → Kinetic energy of turbine → Electrical energy

Advantages

- Higher efficiency than coal plants.
 - Less pollution (no sulphur).
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Electricity Generation & Environment

Why Fossil & Nuclear Fuels are NOT Environment Friendly?

1. Air pollution (CO₂, CO, NO₂, SO₂).
2. Global warming and acid rain.
3. Limited fuel reserves.
4. Nuclear waste & accident risk.

Towards Green Energy

What is Green Energy?

- Energy generated from **renewable and non-polluting sources**.
- Examples: **Water, Wind, Solar, Biofuel**.

Hydroelectric Power Plant

Working

- Water stored in dam has **potential energy**.
- Flowing water has **kinetic energy**.
- Water rotates turbine.
- Turbine rotates generator → electricity produced.

Energy Transformation

Potential energy → Kinetic energy (water) → Kinetic energy (turbine) → Electrical energy

Advantages

1. No air pollution.
2. Renewable source.
3. Water can be reused.

Problems

1. Submergence of land & villages.
2. Affects aquatic life.

Wind Energy

Wind Power Generation

- Wind rotates blades of **wind turbine**.
- Gear box increases speed.

- Generator produces electricity.

Energy Transformation

Kinetic energy of wind → Kinetic energy of turbine → Electrical energy

Advantages

- Clean energy.
- Renewable.

Limitation

- Requires high wind velocity.

Solar Energy

1. Solar Photovoltaic Cell

- Converts sunlight **directly into DC electricity**.
- Based on **photovoltaic effect**.
- Made of **silicon (semiconductor)**.

Series & Parallel Connection

- **Series:** Voltage increases, current same.
- **Parallel:** Current increases, voltage same.

Solar Panel → String → Array

- Cell → Panel → String → Array

Important Points

- Solar power is DC.
 - Inverter converts DC to AC.
 - Batteries store energy.
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2. Solar Thermal Power Plant

- Sunlight converted to **thermal energy**.
 - Steam turbine + generator used.
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